

Investigating Invariances in Auditory Event Categorization With Model Metamers

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Model metamer
sound examples
& CCN-P paper



Carnegie
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Invariance to acoustic features (e.g., pitch and timing) is essential for robust recognition of natural sounds

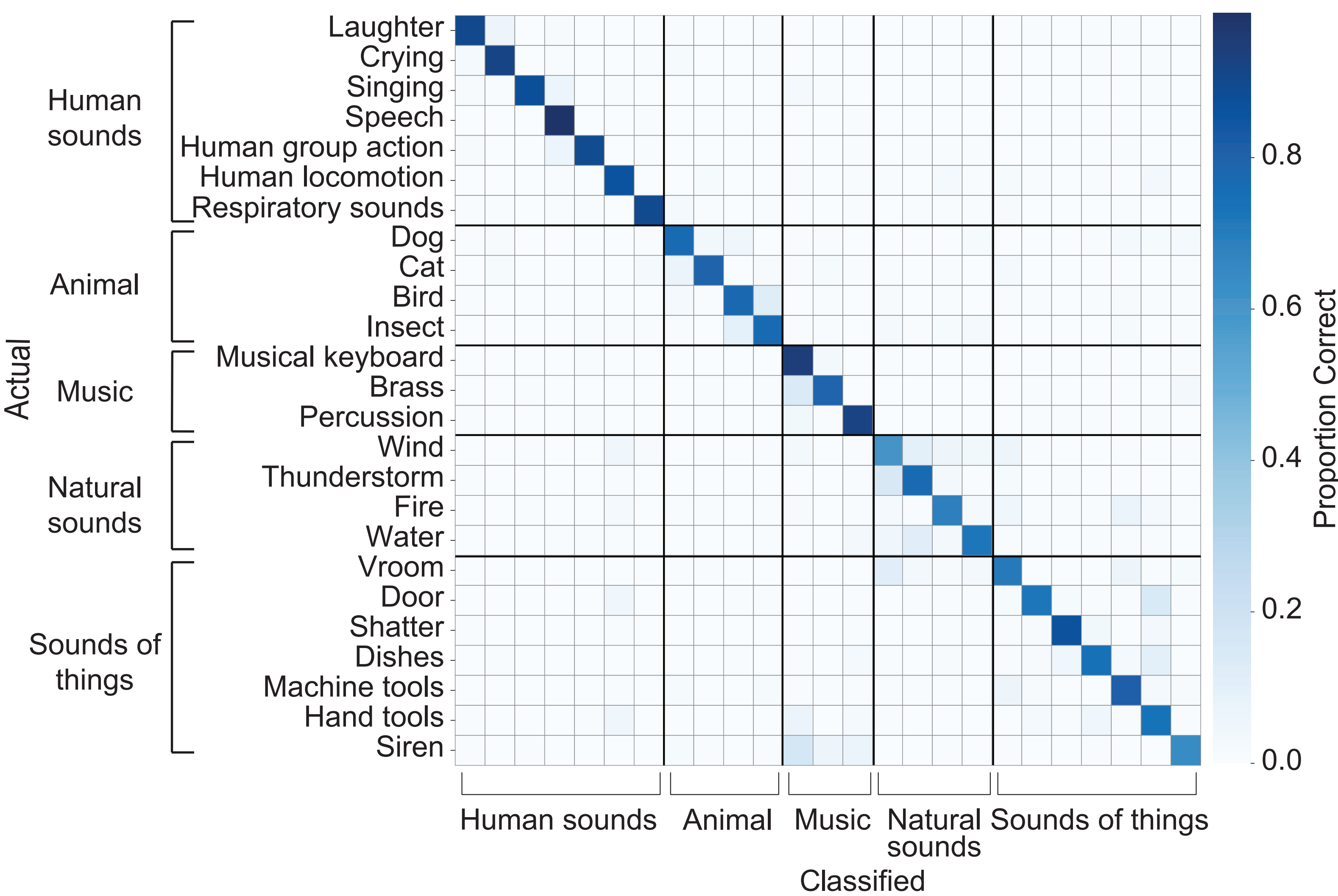
Deep neural networks (DNNs) achieve high performance on auditory event categorization tasks → *Unclear* if DNNs share human-like invariances

Q: Are categorical invariances of DNNs shared with humans?

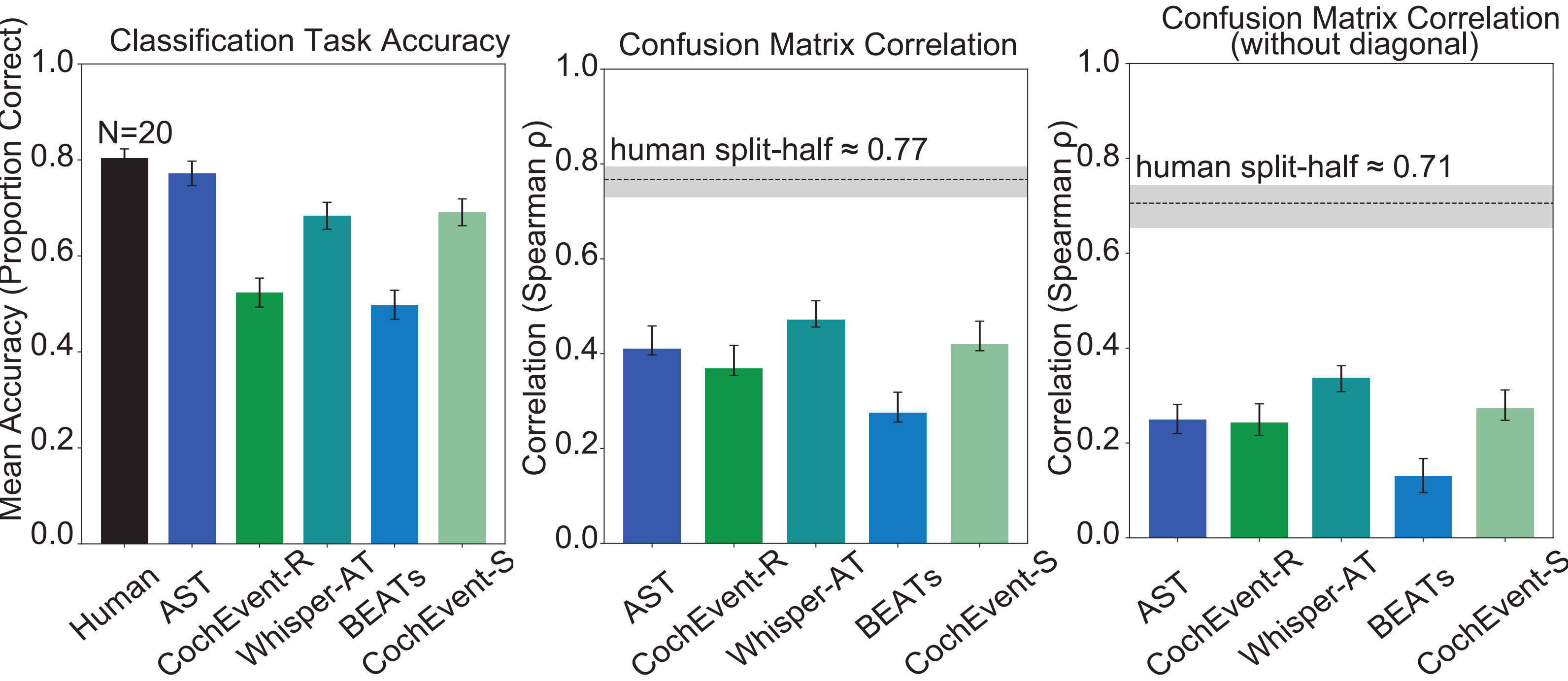
1. Design a behavioral task to study auditory event categorization for natural sounds
2. Compare behavioral patterns of models to humans on the categorization task
3. Use model metamer framework to reveal whether humans and models share similar invariant representations for sound categories

Humans correctly classify natural sounds

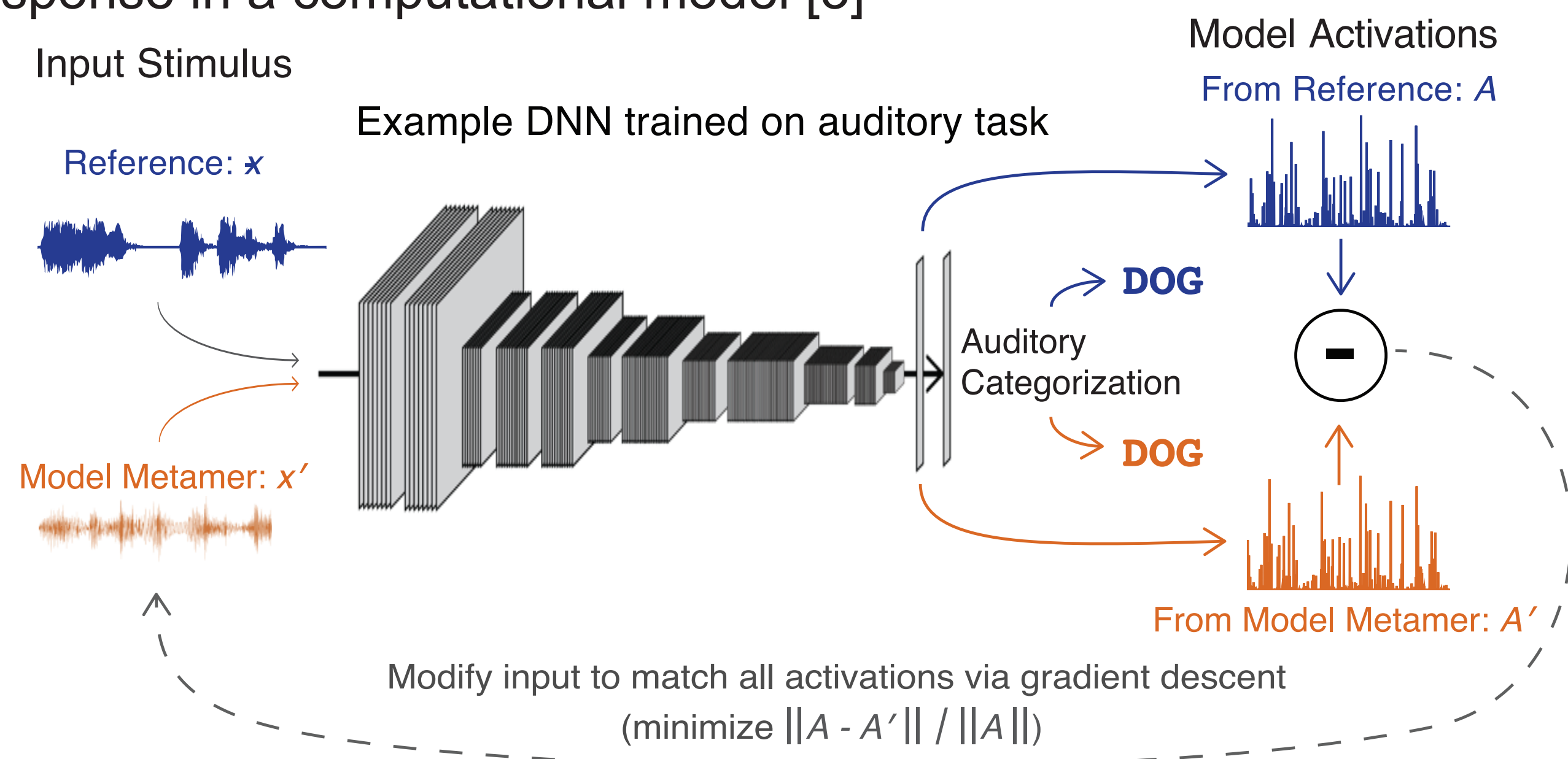
- Developed a 25-way forced choice categorization task, following the AudioSet ontology [1]
- 11 sound clips per category from FSD50K [2]
- All samples were 2s duration
- N=20 participants (Prolific)
- 80% +- 2% accuracy across all categories



Models succeed at classification, but make different errors than humans do



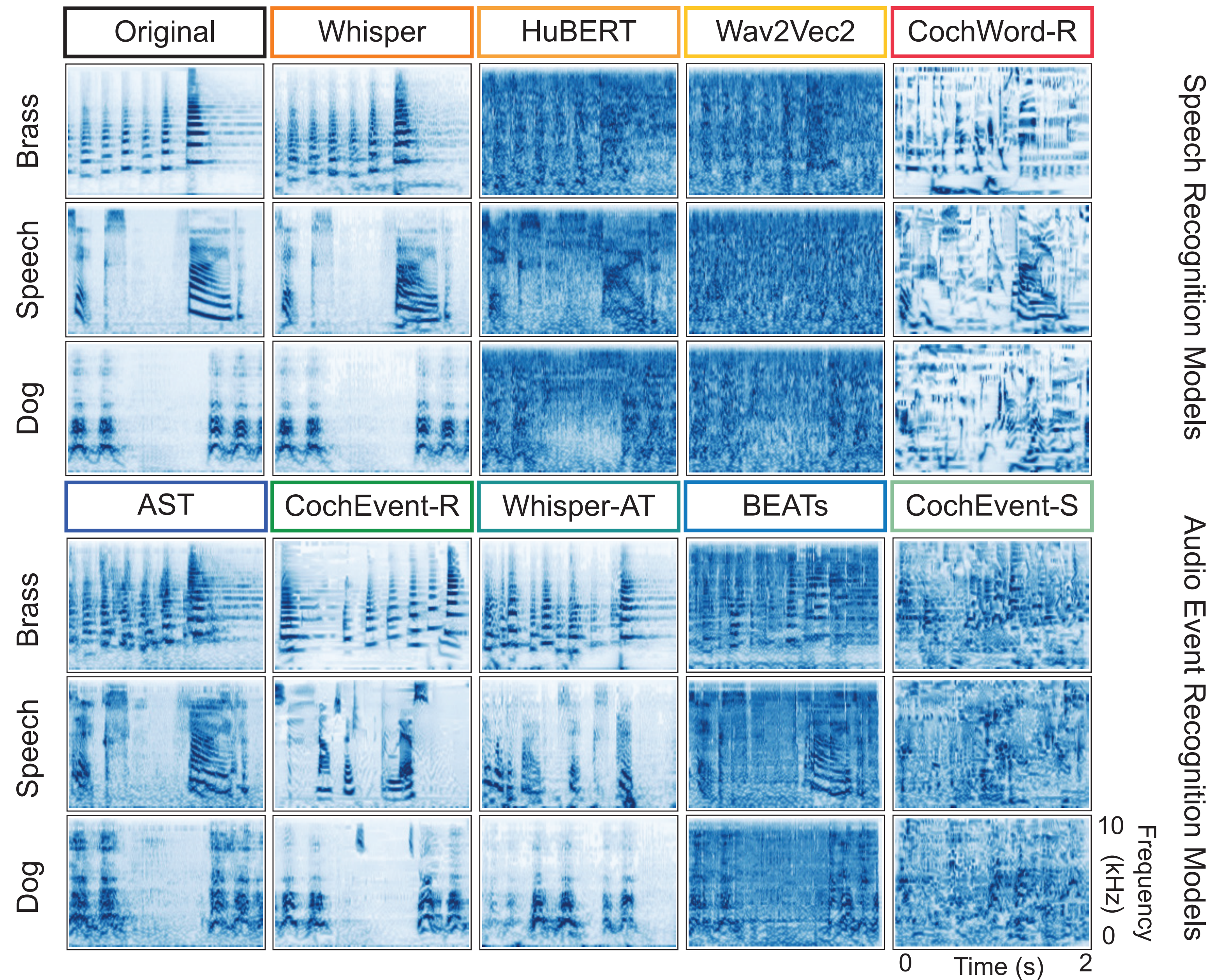
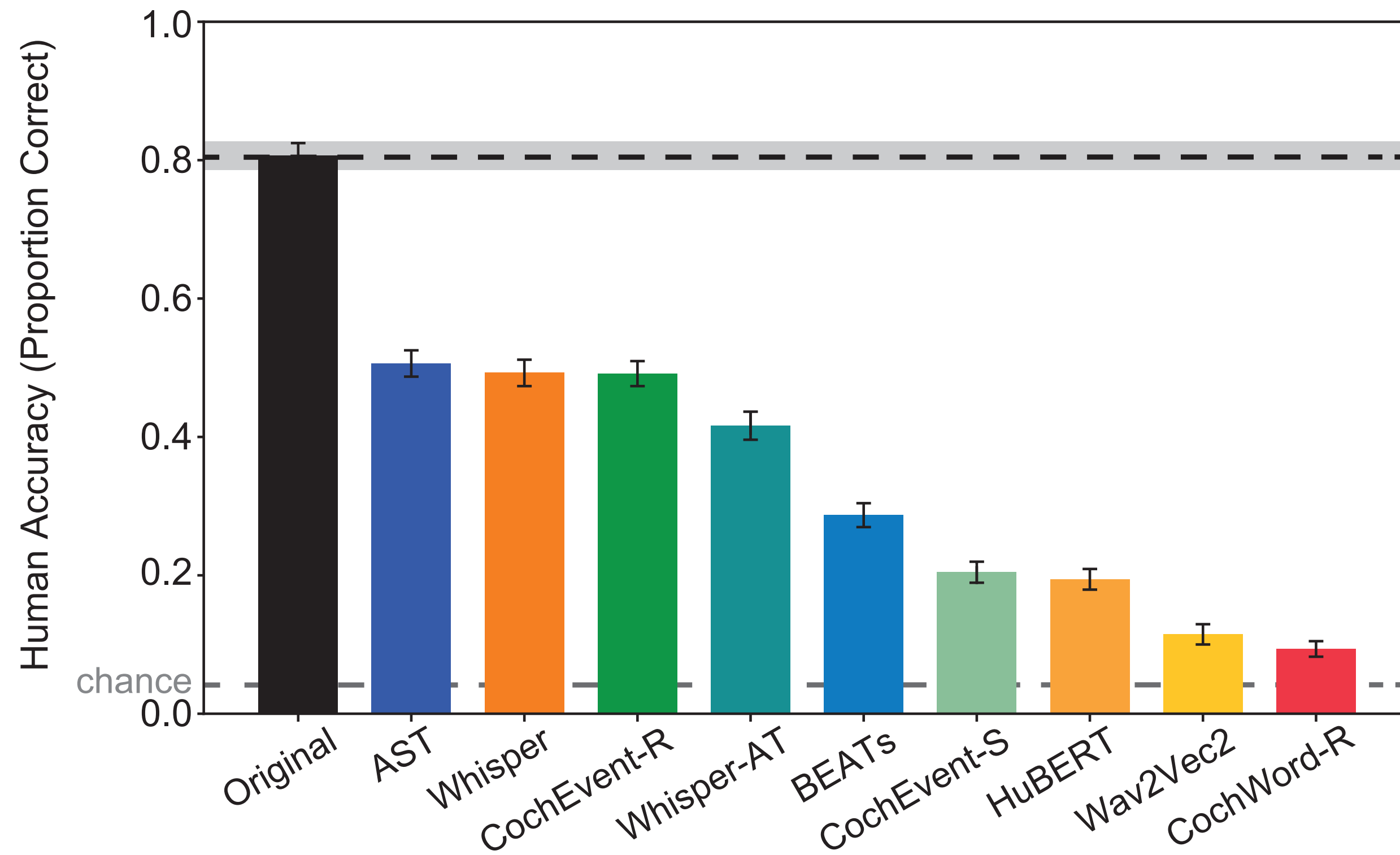
Model metamers: Two stimuli that are physically different but produce the same response in a computational model [3]



Human recognition of model metamers varies with task objective, adversarial model training, and metamer generation layer

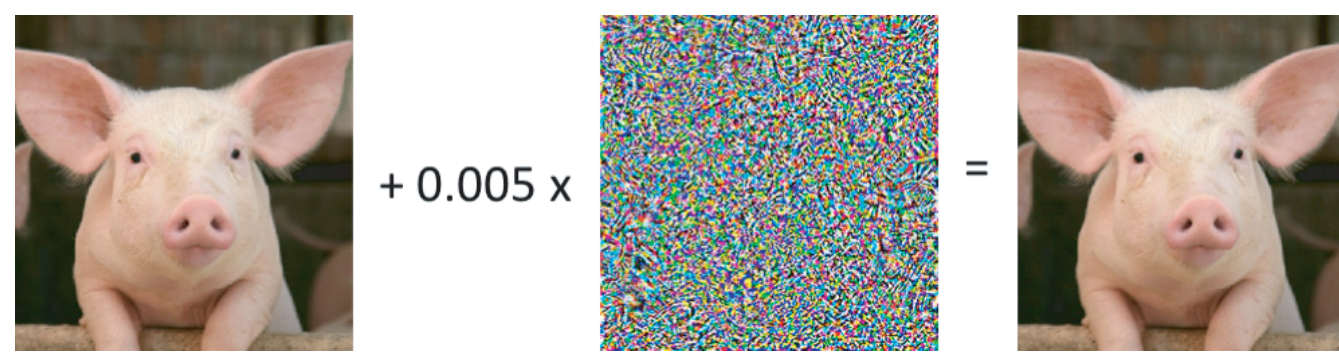
Human recognizability is generally higher for model metamers generated from audio event recognition models than from speech recognition models

- Generated model metamers from the penultimate representation layer for each model and tested human recognizability using the same task
- *Whisper* learns rich representations of non-speech background sounds, even without being explicitly trained for auditory event categorization [4]



Human recognizability of model metamers improves with adversarial training *only* when the model is trained on audio event recognition

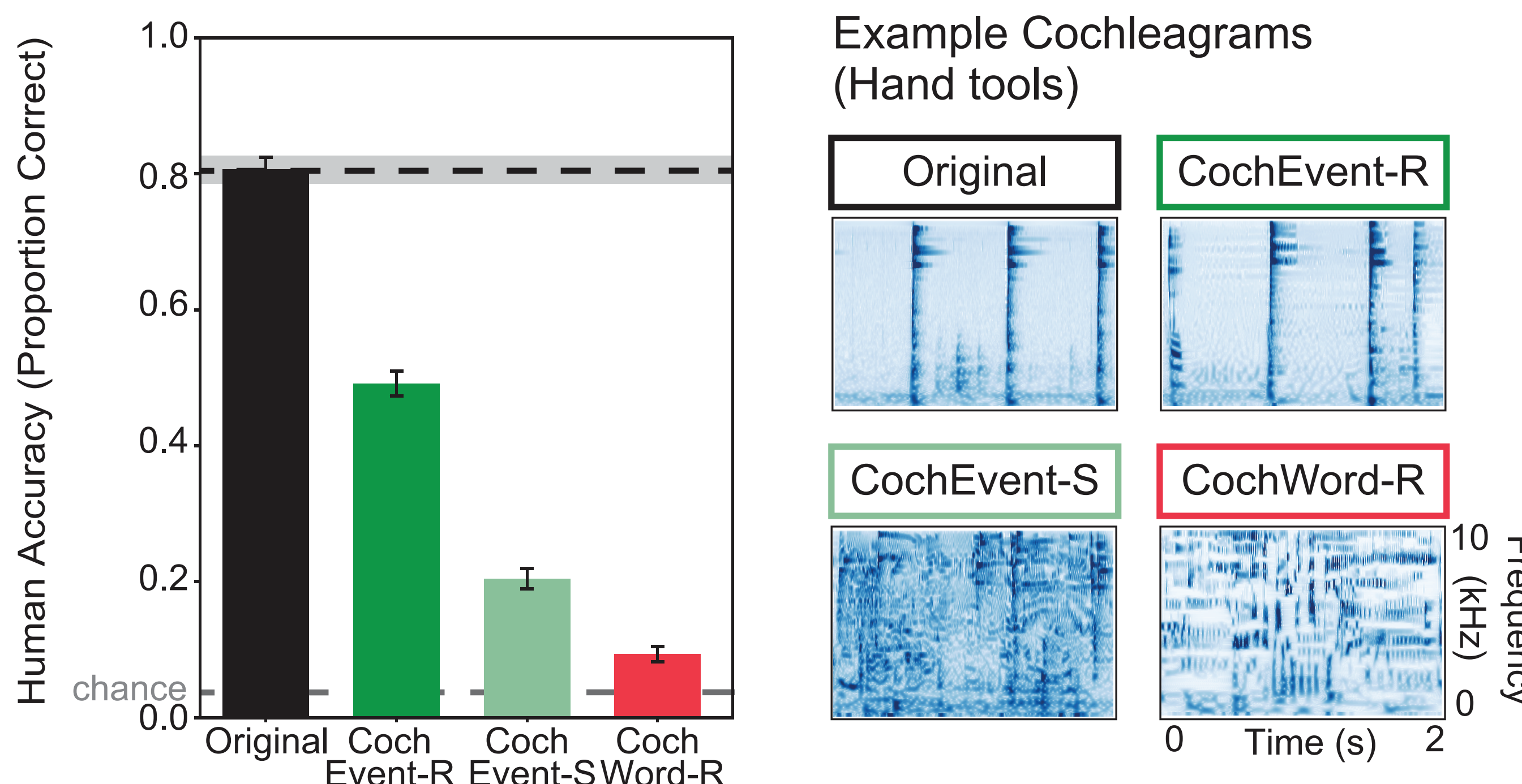
- Adversarial training improves alignment to human perception by increasing robustness to small cochleagram perturbations [3,5]



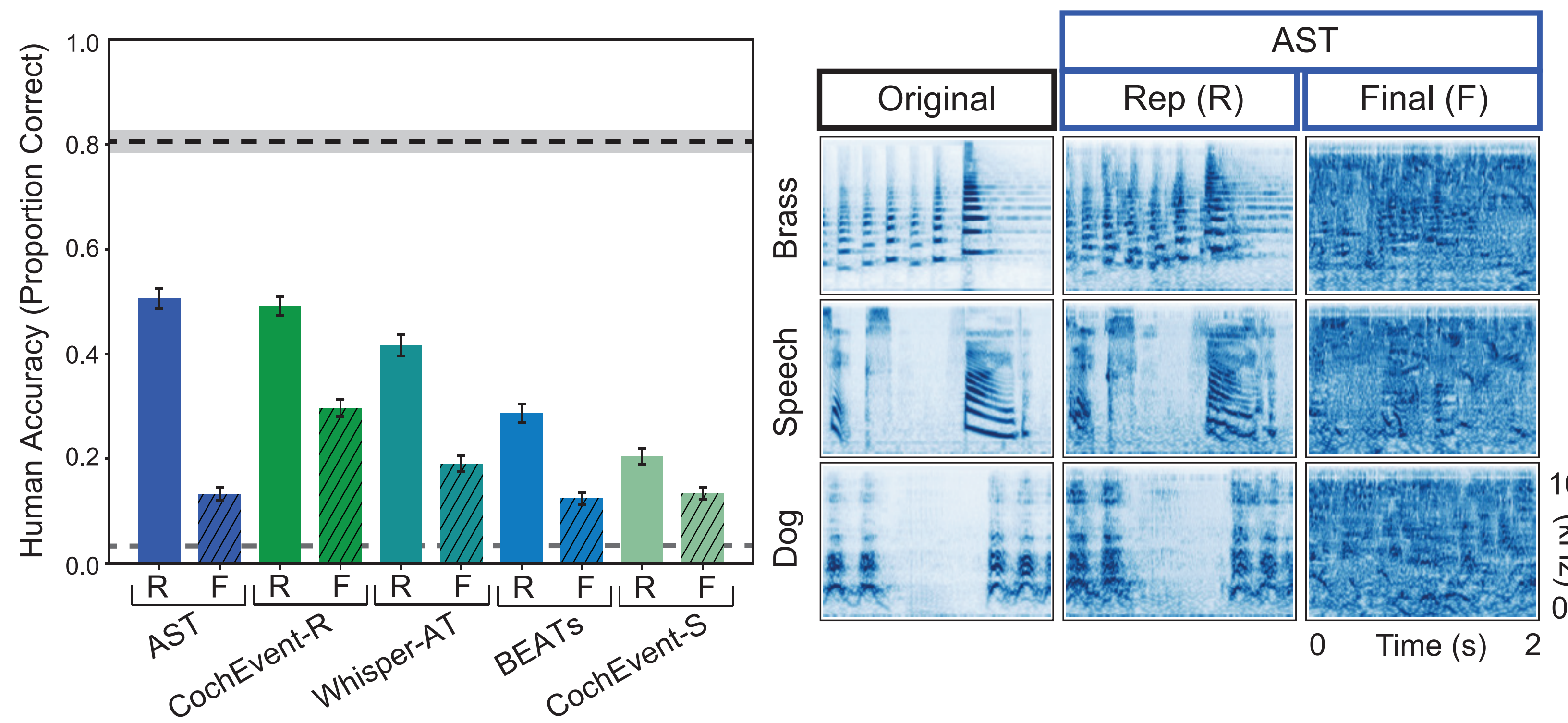
Goodfellow et al. 2015

- Compared a controlled set of models that share the same architecture and training data, but differ in the training task or the training strategy

- **CochEvent-R:** Auditory event recognition model with adversarial training
- **CochEvent-S:** Auditory event recognition model with standard training
- **CochWord-R:** Speech recognition model with adversarial training

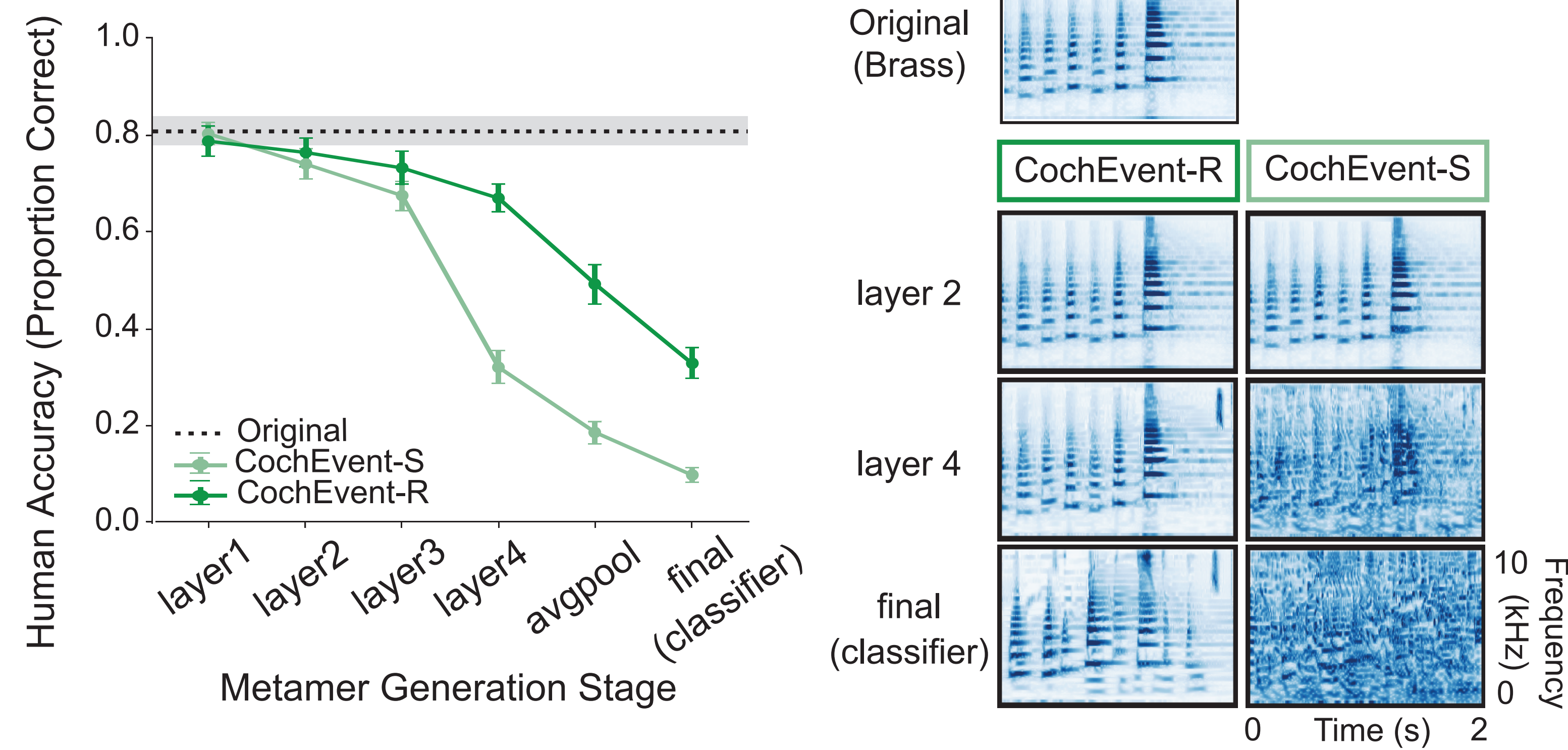


Human recognizability of model metamers significantly drops at the final classification layer

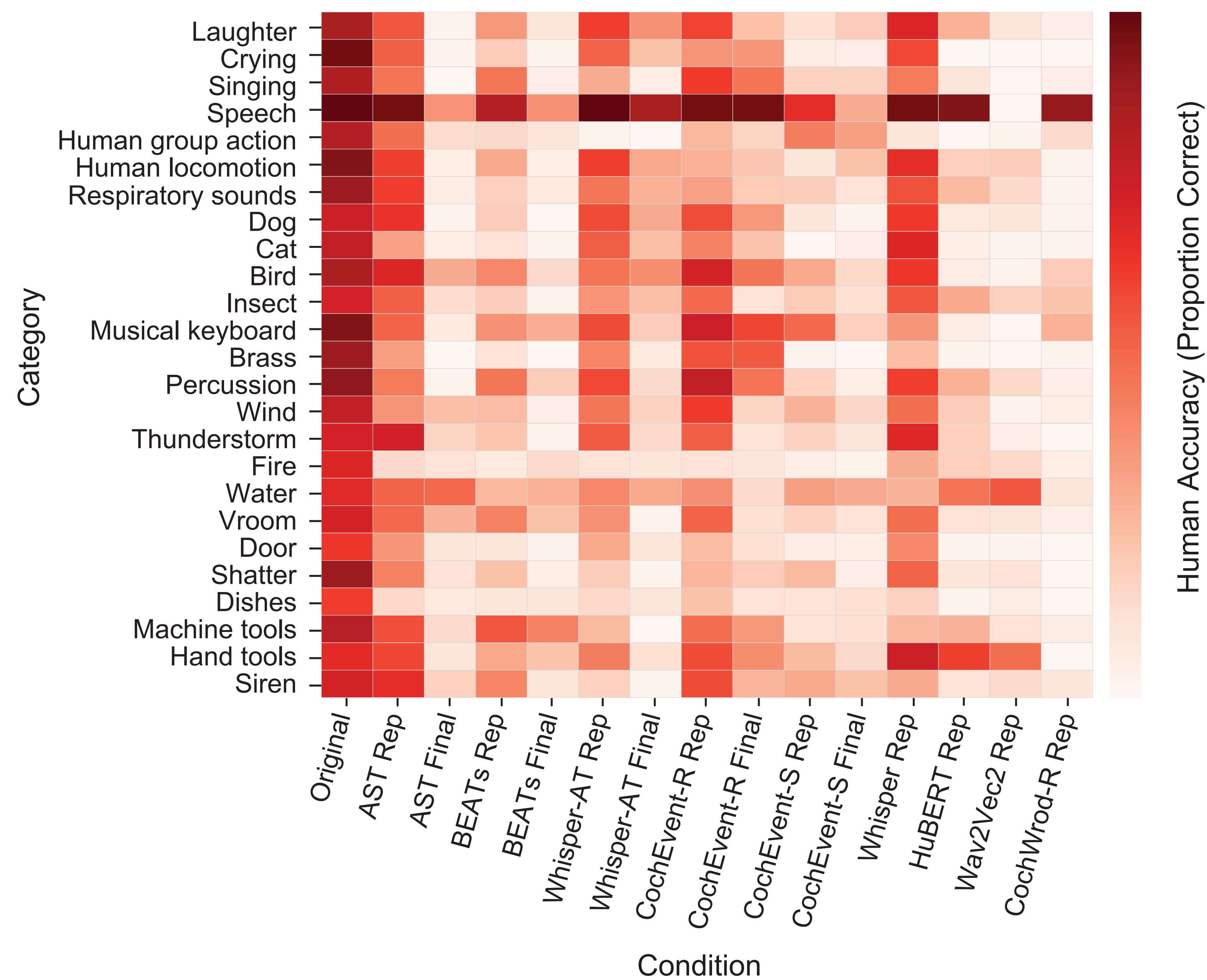


Intermediate model stages of standard and adversarially trained models show where model invariances diverge

- Compared two models that only differ in the training strategy but share the same architecture and training dataset



Human recognizability of model metamers exhibits distinct category-specific patterns across model conditions



Conclusion

1. Developed a paradigm testing model-human invariance alignment in audio event categorization
2. DNNs show high classification accuracy but make different errors than humans do
3. Metamer recognizability increases with task specificity and adversarial training
 - Models exhibit non-human-like invariances at the final classification stage
 - Metamers from intermediate stages show where model invariances diverge

Funding: This work is supported in part by MURI N00014-19-12332

References: [1] Gemmeke et al. 2017 [2] Fonseca et al. 2021 [3] Feather et al. 2023 [4] Gong et al. 2023 [5] Goodfellow et al. 2015